

LESSON GUIDE

Technology Plus

Unit 7: Security

6.4.1 – Plain Text vs. Cipher Text

Lesson Overview:

In this lesson, students will differentiate between plain text and cipher text and explore how encryption keeps data safe and secure.

Students will:

- Describe the process of encryption and decryption.
- Identify and describe plain text and cipher text.
- Apply Caesar Cipher to encrypt and decrypt secret messages.

Guiding Question: OVD JHU LUJYFWAPVU RLLW KHAH WYPCHAL HUK ZLJBYL?

Suggested Grade Levels: 8-12

Technology Needed: Yes

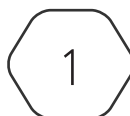
Approximate Time Required: 50-65 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 6.4 – Identify common use cases for encryption.
 - **Plain text vs. cipher text**
 - Data at rest
 - File level
 - Disk level
 - Mobile device
 - Data in transit
 - Email
 - HTTPS
 - VPN
 - Mobile application

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Lesson Background

Background Information:

This lesson introduces students to the foundational concept of encryption, focusing specifically on the difference between plain text and cipher text. Encryption is the process of transforming readable information (plain text) into an encoded format (cipher text) to protect it from unauthorized access. Understanding this concept helps students grasp how digital information—such as text messages, emails, and passwords—is kept secure in everyday life. The lesson also connects to historical examples, like the Caesar Cipher, to demonstrate how encryption has been used for centuries to protect sensitive information. Teachers should emphasize real-world applications, including protecting personal data, secure communication, and online privacy, so students can see the relevance of encryption in their daily digital lives.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 6.4.1 – Plain Text vs. Cipher Text • Activity 6.4.1– Secret Agent Shift
Materials per Student	• Student Notes 6.4.1 – Plain Text vs. Cipher Text • Student Handout 6.4.1 – Plain Text vs. Cipher Text • Assessment 6.4.1 – Plain Text vs. Cipher Text (optional) • Cipher Disk Printout (optional)

Cyber Terms:

Term	Definition
encryption	the process of encoding information or data into a form that is unintelligible
key	a unique value that tells an encryption algorithm how to encrypt or decrypt information or data
decryption	the process of turning unreadable information or data back into a readable format
algorithm	a set of rules and instructions that a computer follows to solve problems or make decisions
plain text	data or information that is in its original, readable format
cipher text	the unreadable result of applying an encryption algorithm to plain text

6.4.1 – Plain Text vs. Cipher Text

Guiding Question: OVD JHU LUJYFWAPVU RLLW KHAH WYPCHAL HUK ZLJBYL?

Lesson Launch (5 minutes)

- **Ask** students to imagine: “You write a secret note to a friend across the classroom. What if someone else grabs the note and reads it? How could you keep the contents safe?”
- Give students time to think and invite a few to share ideas.
- Accept all reasonable answers (folding, codes, hiding it, locking it up, etc.), then connect back: “These are all ways to protect information – encryption works the same way for digital messages.”
- Students will uncover the meaning of the guiding question by the end of the lesson.

Plain Text, Cipher Secrets (20-25 minutes)

- Distribute the Student Handout 6.4.1 – Plain Text vs. Cipher Text and have students fill in the answers during the lesson.
- Introduce students to **encryption**, its definition, and benefits.
 - Introduce a **key** and **decryption**.
 - Encryption is beneficial because it:
 - Protects Privacy: it keeps personal information safe from people who shouldn’t see it
 - Keeps Information Secure: stops malicious actors from stealing sensitive data
 - Builds Trust: Sites with HTTPS encrypt your connection, protecting data sent between your browser and the website
 - Prevents Misuse: even if someone gets encrypted data, it is useless without the key
- Briefly share examples of early encryption (Caesar Cipher, substitution, transposition).
- Explain that modern encryption uses algorithms – step-by-step instructions that define how plain text is transformed into cipher text.
 - An **algorithm** is a step-by-step set of instructions for solving a problem or completing a task.
 - Algorithms define how the data is scrambled in an encryption or how plain text is transformed into cipher text.
- Introduce **plain text** and discuss its usage.
- Introduce **cipher text** and discuss its usage.
- **Caesar Cipher**, also known as a shift cipher, is a simple encryption method where each letter in

plain text is shifted a certain number of places down the alphabet to create cipher text.

- Share the encryption example.
 - Share the decryption example.
 - Today, students will use Caesar Cipher to encrypt and decrypt messages.
 - Caesar Cipher is one of the oldest and simplest methods of encryption – it shows the basic idea of scrambling data, which modern algorithms build on.
- Students will learn more about encryption in lesson 6.4.2 – Data Types.

Activity: Secret Agent Shift (20-30 minutes)

- Display Activity Slides 6.4.1 – Secret Agent Shift, have students record their answers in the handout.
- For this activity, students will use our Caesar Cipher Tool <https://tools.cyber.org/caesar> in a browser.
- If for some reason students cannot access the Caesar Cipher Tool in their web browser, have them login to the CYBER.ORG Range and use the Ubuntu Machine to access the tool from there.
- Another alternative method or bonus for this activity is to have students create and use their own Cipher Disk. At the following link, you will find the PDF and instructions for creating a Cipher Disk. This disk can be used, just like the Cipher Tool, to decrypt and encrypt messages. <https://www.cyber.org/cipher>
- Students will use the tool to encrypt six plain text messages into cipher text, decrypt six cipher text messages into plain text, decrypt today's guiding question, and write a cipher message for you.
- Allow students to read the description on the Caesar Cipher page before starting the activity.
- All messages typed into the tool will show and return in all capital letters.
- When students create their own messages in cipher text, remind them to keep them school appropriate.

Putting It All Together (5 minutes)

Discussion Questions

- Review what students learned today by discussing the questions below as a class.
- What is encryption and why is it beneficial?

Encryption is the process of converting information or data into an unreadable form to prevent unauthorized access. It protects privacy, keeps information secure, builds trust, and prevents misuse of data.

- What is plain text and what is it used for?

Plain text is original, readable data. It is used for direct communication, documentation, and any situation where no security is applied.

- What is cipher text and what is it used for?

Cipher text is the result of applying an encryption algorithm to plain text; it is the scrambled, unreadable version of the original data. It is used to secure data during transmission or storage, keeping it confidential and safe from unauthorized access.

- What was today's guiding question?

How can encryption keep data private and secure?